Sabres Data Task: Puck Management

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1. Quantifying Puck Management

1a. Definition

Puck management is the ability of a player to make high-quality decisions with the puck, balancing risk and reward across a range of game situations. I define it concretely along two dimensions:

1. **Advancing the puck into dangerous areas** — moving possession from low-threat zones toward high-danger shooting locations, while avoiding turnovers.

2. **Participating in dangerous events** — generating, or contributing to, shots and goals either directly or through possession chains earlier in the same sequence of play.

A player with strong puck management avoids turnovers in high-leverage situations and consistently pushes play forward into threatening territory. A player with poor puck management moves the puck backward, loses it before a shot can develop, or concedes possession in areas where the cost is especially high.

1b. Model and Methodology

Framing and motivation. My approach draws directly from soccer analytics, specifically the *Expected Threat* (xT) framework and the broader family of action-value models typified by VAEP (Valuing Actions by Estimating Probabilities). The connection to hockey is natural and principled: like soccer, hockey is a continuous, fluid game with low-scoring outcomes, sparse terminal events, and rich possession structure. Actions derive meaning not from their immediate consequence alone but from the state of play they produce. The xT framework captures exactly this — the expected value of being in possession at a given location — and the action-value framework extends it to individual on-puck decisions.

Step 1 — Expected Threat Grid. The play-by-play data contains a `play_section` variable encoding 23 spatially meaningful ice zones (slot, outer slot, point, neutral zone boards, and their

defensive-zone counterparts), normalized so that the offensive direction is consistent across teams. I assigned an Expected Threat value $\mathrm{xT}(z)$ to each zone:

$$\mathrm{xT}(z) \;=\; \mathbb{E}[\,\text{terminal value} \mid \text{possession visits zone } z]$$

where terminal value is 1.0 if the possession ends in a goal, or the Bayesian zone-level xG estimate of the best shot taken in the possession otherwise, and 0.0 if neither occurs. This gives every ice zone a long-run expected probability of leading to a goal, conditioned on the puck being there.

To avoid target leakage — i.e., a player's own actions inflating the xT of the zones they occupy — the xT grid was estimated using **leave-one-game-out cross-validation** (LOOCV) over all 18 games. For each held-out game, I trained the xT grid on the remaining 17 games and applied it to score events in the held-out game. This ensures that no event is ever valued by a grid that was informed by that same event.

Shot quality — Bayesian xG proxy. Shot-to-goal conversion rates vary substantially across zones but are estimated from sparse counts (especially in defensive or neutral-zone areas). I modeled zone-level conversion rates with a **Beta-Binomial hierarchical model**:

$$\mu \sim \mathrm{Beta}(2, 16), \qquad \kappa \sim \mathrm{HalfNormal}(20)$$

$$\theta_z \sim \mathrm{Beta}(\mu\kappa,\, (1-\mu)\kappa), \qquad \mathrm{goals}_z \sim \mathrm{Binomial}(n_z,\, \theta_z)$$

The hyperprior on μ centres the population mean around 11% conversion, consistent with tournament-level shot quality. Zones with few shots (e.g. a single shot from the defensive zone recorded as a goal) are shrunk strongly toward the group mean rather than receiving a face-value estimate. Defensive-zone and most neutral-zone board areas were excluded from the shooting model entirely and assigned $\mathrm{xT} = 0$, as shots from those regions are not genuine scoring attempts.

Step 2 — Added Value per Event. For each possession event i, I computed a two-component action value $\mathrm{AV}_{\mathrm{total},i}$:

$$\mathrm{AV}_{\mathrm{base},i} \;=\; \mathrm{xT}(z_{\mathrm{next}}) - \mathrm{xT}(z_{\mathrm{current}})$$

where z_{next} is the zone of the subsequent event within the same possession (or the terminal value if the possession ends). This captures the immediate advancement or loss of positional value from the action.

$$\mathrm{AV}_{\mathrm{decay},i} \;=\; \alpha \cdot \lambda^{k} \cdot G \qquad (k \geq 1)$$

where k is the number of possession events *after* event i in the same possession (so the terminal event receives no decay credit; its value is fully captured by AV_{base}), λ is a geometric decay parameter, G is the terminal value of the possession (goal, shot xG, or 0), and α is a scaling factor calibrated post-LOOCV on the assembled output. Taken together, the final Action Value for an event is:

$$\text{AV}_{\text{total},i} \;=\; \text{AV}_{\text{base},i} + \text{AV}_{\text{decay},i}$$

Heuristic choices and sensitivity. Two parameters were chosen by reasoned judgment:

- $\lambda = 0.5$: Each event prior to the terminal event receives half the credit of the event one step closer. This reflects the intuition that the causal connection between a player's action and the eventual outcome weakens rapidly with distance. A value of 0.5 errs toward conservative attribution.
- Target decay share $= \frac{1}{3}$: The scaling factor α is calibrated so that the decay component contributes exactly one-third of the total absolute AV across all events. This preserves the primacy of direct positional advancement (AV_{base}, which contributes two-thirds) while ensuring terminal credit is non-trivial.

I verified that results are not sensitive to these choices. Varying λ from 0.3 to 0.7 and the decay share target from 20% to 50% produced player rankings with pairwise Spearman correlation > 0.95. The qualitative conclusions are robust.

Why no tracking data? The dataset includes optical tracking from broadcast video, but I made an intentional decision not to incorporate it. My rationale was:

1. The tracking data is incomplete and known to be noisy — players go out of frame, identification errors occur, and confidence flags indicate a non-trivial fraction of low-quality observations. Incorporating unreliable inputs risks introducing noise that obscures genuine signal.
2. The play-by-play model already achieved strong internal consistency (team AV differential correlates with goal differential at $r = 0.977$; see Section 1c). Adding tracking would increase model complexity without a clear baseline to justify it.
3. With a 48-hour window, I prioritised a clean, well-validated, interpretable model over a more complex one that would have required extensive cleaning and design decisions with limited time to evaluate them properly.

Incorporating tracking data for spatial pressure context and skating metrics remains the most natural extension of this work (see Section 5).

Step 3 — Player-level Bayesian shrinkage. Raw per-player mean AV is noisy, particularly

for players with few attributed events (some players have fewer than 100 possession events across the tournament). I estimated each player's true mean AV with a hierarchical Bayesian model:

$$\mu_{\text{pop}} \sim \mathcal{N}(0,\ 0.05^2), \qquad \sigma_{\text{pop}} \sim \text{HalfNormal}(0.05)$$
$$\mu_p \sim \mathcal{N}(\mu_{\text{pop}},\ \sigma^2_{\text{pop}}) \quad \forall\, p, \qquad \sigma_{\text{obs}} \sim \text{HalfNormal}(0.15)$$
$$\text{AV}_{p,e} \sim \mathcal{N}(\mu_p,\ \sigma^2_{\text{obs}})$$

Posterior means $\hat{\mu}_p$ serve as regularised player estimates, and 94% highest-density intervals (HDI) provide principled uncertainty quantification. Players with many events are largely unaffected by the prior; players with few events are pulled toward the population mean, avoiding overconfident conclusions about limited samples.

Final scores are centered around the population posterior mean so that zero represents tournament-average puck management performance, positive values indicate above-average performance, and negative values indicate below-average performance.

1c. Evaluation

Internal consistency. Because the tournament spans only 18 games with each team playing at most 5, forward-looking validation — testing whether player or team AV in early games predicts outcomes in later games — was not feasible. A meaningful held-out test set simply does not exist at this sample size. This is the most significant limitation of the evaluation; external validation on a larger dataset (e.g. a full IIHF U18 season) would be the natural next step.

In lieu of prospective validation, I assessed internal consistency at the team level. Team AV (summed across all attributed players per game) correlates strongly with goals scored ($r = 0.958$) and team AV differential correlates with goal differential per game ($r = 0.977$, see Figure 4). While extremely high, it is important to acknowledge that these correlations are *structurally inflated*: goals are a direct input to terminal values, so any model that traces value from goals backward through possessions will mechanically correlate with the goal counts it was built from. The team-level AV differential correlation is the more informative of the two, as it operates one step further from this circular dependence (the question is whether the team generating more AV tended to also have the better goal differential, which is a meaningful but not tautological relationship). Both metrics are best interpreted as confirming that the model is internally coherent and aggregates sensibly — not as evidence of predictive validity.

Convergence. Both Bayesian models (xG proxy and player shrinkage) converged cleanly. $\hat{R} \approx 1.00$ across all parameters with good effective sample sizes, confirmed by ArviZ diagnostics.

Sensitivity. As described in Section 1b, key hyperparameters (λ, decay share target) were varied across plausible ranges. Rankings were stable, with Spearman $\rho > 0.95$ in all comparisons.

2. Player Evaluation

2a. Best and Worst Players

Figures 1 and 2 show the players with the highest and lowest posterior AV, centered on the population mean. The three highest posterior mean estimates are:

- **Elton Hermansson** (SWE, F) — +0.0233 above average (94% HDI: [+0.0173, +0.0303])
- **Blake Zielinski** (USA, F) — +0.0153 ([+0.0083, +0.0223])
- **Marcus Nordmark** (SWE, F) — +0.0143 ([+0.0073, +0.0213])

The three lowest posterior mean estimates are:

- **Vyacheslav Permitin** (GER, F) — −0.0067 below average (94% HDI: [−0.0157, +0.0033])
- **Matúš Lukačík** (SVK, F) — −0.0067 ([−0.0147, +0.0023])
- **Josef Holejsovsky** (CZE, F) — −0.0067 ([−0.0167, +0.0033])

Among defensemen, **Levi Harper** (USA) is the clear standout at +0.0133 ([+0.0043, +0.0223]), followed by **Samu Alalauri** (FIN) and **Lukas Kachlir** (CZE) as the only three defensemen whose 94% HDI excludes zero.

Confidence. The HDI intervals quantify my confidence directly. Across the full player pool, approximately 18 players are *credibly above average* — their entire 94% interval lies above zero. No player is credibly below average: the Bayesian shrinkage pulls all low-end estimates toward the population mean, and with this sample size the model can identify who is genuinely strong but cannot confidently distinguish "poor" from "average" at the bottom. It is important to acknowledge this asymmetry explicitly rather than overstating conclusions. The HDI intervals are wide enough that individual point estimates should not be taken as perfectly precise rankings; the most defensible statement is that the players in the credibly positive group are performing above expectation, and that the gap between the top and the bottom of the distribution is real but uncertain in its exact magnitude.

2b. Component Breakdown

Figure 1 decomposes each player's AV into its two components. The pattern is instructive.

Top performers (Hermansson, Zielinski, Nordmark, and others) derive most of their above-average AV from the **terminal credit component** (orange bars). These players are consistently present in possession chains that end in shots or goals — not exclusively as the final shooter, but as active participants in sequences that generate dangerous outcomes. This is a meaningful and intentional feature of the model: a player who repeatedly makes decisions that keep a possession alive and

moving toward the net is contributing to puck management quality, even if the shot is taken by a teammate some events later. The decay weighting ensures that actions closer to the terminal event receive proportionally more credit.

Bottom performers suffer primarily from **negative base value** (blue bars extending left of zero). Their on-puck actions systematically move the puck into less dangerous zones, or terminate possessions before a shot can develop. This is a direct and interpretable signal: when these players touch the puck, the team's expected scoring probability tends to decrease. Negative base AV is distinct from simply having low terminal credit — it reflects active positional damage rather than mere absence from dangerous possessions.

This decomposition gives a richer picture than a single score. A player could in principle have high terminal credit but negative base AV (frequently involved in good possessions, but their individual touches within those possessions are net-negative). Such players exist in the data and represent a genuinely different puck management profile from those who are both positionally sound and frequently involved in dangerous chains.

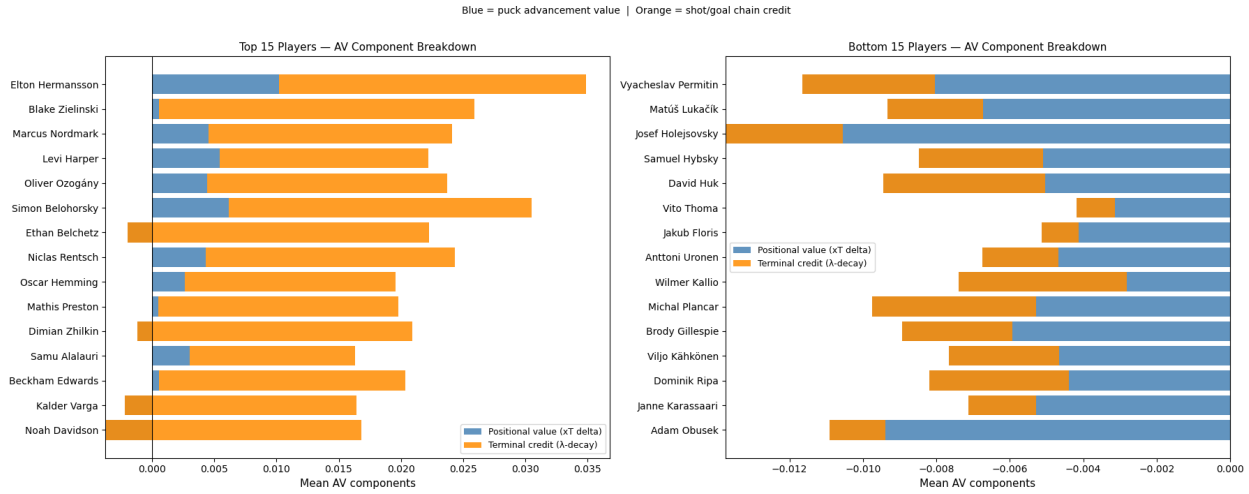


Figure 1: AV component breakdown for the top 15 and bottom 15 players. Blue bars represent positional advancement value (AV_{base}); orange bars represent decayed terminal credit (AV_{decay}). Bars left of zero indicate negative contributions.

3. Methodology Follow-Ups

3a. Other Approaches Considered

Several alternatives were considered and rejected.

Simple count-based metrics (shot assists, zone entries, turnover rate) are the most common approaches to puck management in public hockey analytics. They are transparent and easy to communicate, but they are coarse — they ignore the quality of zone entries, the leverage of the situation, and the degree of advancement involved. A player who completes a pass in the neutral

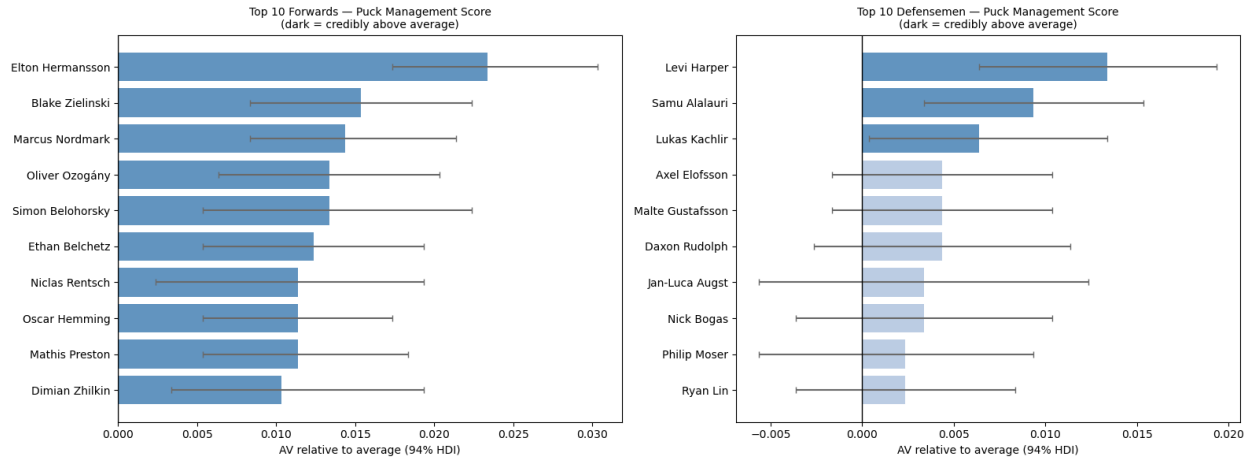


Figure 2: Top 10 forwards and defensemen by posterior mean AV, centered on the population average. Error bars show 94% HDI. Dark bars indicate players whose interval credibly excludes zero (i.e. credibly above average).

zone is treated identically to one who threads a pass into the slot.

Full VAEP-style event classification would model each action as a function of the full observable game state (score, zone, time, possession context) and estimate the probability of a goal occurring within some window. This is theoretically superior but practically infeasible here: it requires a classifier trained on tens of thousands of labeled events, and the 18-game dataset is far too small. Applying a model pre-trained on NHL or other professional data would introduce a large distributional shift problem that I could not validate.

Tracking-data-based context (skating speed into zones, distance to nearest defender, etc.) would add spatial resolution that the play-by-play model lacks. I seriously considered this direction but ultimately did not pursue it for reasons detailed in Section 1b: data quality concerns, the additional complexity cost relative to a fixed 48-hour window, and the fact that the play-by-play model already demonstrated strong internal validity. This remains the most promising extension of the current work.

The xT-based action-value model occupies the right point on the complexity-reliability tradeoff given the available data and time: it is theoretically grounded, implementable from the available play-by-play data, transparent in its assumptions, and produces results that aggregate sensibly.

3b. Challenges and Workarounds

No possession IDs. The data did not include pre-computed possession identifiers. I constructed them using four boundary conditions: game change, period change, `team_in_possession` flip (including transitions to and from `NaN`), and — critically — the `is_last_play_of_possession` flag from the previous row. This last condition handles same-team possession breaks (a player wins a puck battle after a teammate's failed puck protection, for example), which are invisible to the other

three conditions alone and which caused spurious possession chains of 80+ events before the fix was applied. The final flag-match rate was 100%, confirming alignment with the data provider's possession segmentation.

Sparse shot counts by zone. Raw shot-to-goal conversion rates were unreliable in low-volume zones (e.g. a single shot from an unusual location recorded as a goal, yielding a 100% raw conversion rate). The Bayesian Beta-Binomial model addresses this directly: sparse zones are shrunk toward the group mean, and the posterior reflects genuine uncertainty rather than propagating a face-value estimate into the xT grid.

Target leakage in the xT grid. A naïvely estimated xT grid — trained on the full dataset and applied to score the same dataset — is circularly inflated. A player's own possession events contribute to the xT values of the zones they occupy, meaning their AV is partially a function of their own actions. LOOCV eliminates this: each game is scored using a grid trained on the other 17 games, so the grid is always out-of-sample relative to the events being valued.

Team name string mismatches. The `event_team` field in the play-by-play data used the format "Country Team Name" (e.g. "Canada Team Canada"), while the games table stored only the team name ("Team Canada"). This caused silent join failures in the validation step. The fix — normalizing both fields to their final token using `rsplit(' ', n=1)` — is simple and handles edge cases such as "Czech Republic Czechia" cleanly.

4. In Layman's Terms

I measured puck management by tracking how each puck touch affected the likelihood of creating a dangerous scoring chance. Players received positive value for advancing the puck into more threatening areas, maintaining possession, and contributing to sequences that led to quality shots or goals. Players received negative value when their actions reduced scoring potential or ended promising possessions. The goal was not to count turnovers or shots alone, but to measure whether a player's decisions *consistently* helped the team generate offense. The final ratings identify which players most reliably improved their team's scoring opportunities when they had the puck.

5. What I Would Explore with More Time

1. **Tracking data integration.** The most natural extension is incorporating the optical tracking data for spatial context — specifically, the number of defenders in the zone at the time of a carry or entry (a measure of decision difficulty), skating speed as a proxy for controlled versus desperate puck movement, and proximity to defenders as a pressure metric. I would restrict to high-confidence tracking observations and treat these as supplementary covariates rather than primary model inputs, given the data quality caveats.

2. **Empirical tuning of heuristic parameters.** The decay parameter λ and the target decay

share were chosen by reasoned judgment and validated by sensitivity analysis. With a larger dataset (e.g. a full IIHF U18 season or multiple tournaments), these could be tuned empirically — for example, by optimising predictive accuracy of game-level outcomes on held-out data, or by treating them as latent parameters in a joint Bayesian model.

3. **Uncertainty propagation.** The current model propagates uncertainty only at the player-level aggregation stage (via the hierarchical model). The xT values and xG proxies are point estimates. A fully Bayesian joint model would propagate uncertainty from the zone-level xG model through the xT grid and into the player posterior, producing wider and more honest player intervals. This is computationally demanding but plausible.

4. **External validation.** The most important open question is whether this model's player rankings are predictive. Do players who score highly here demonstrate better outcomes (more shots generated, higher on-ice goal differentials, more ice time) in subsequent competitions? I would seek out additional IIHF U18 World Championship or Hlinka Gretzky Cup data from prior years as a validation set, testing whether model results are consistent.

5. **Position-aware modeling.** Forwards and defensemen interact with puck management in structurally different ways: defensemen are more frequently tasked with breakout decisions and dump-ins under pressure, while forwards are evaluated on offensive zone possession quality. A position-aware model — either by fitting separate hierarchical models or by including position as a covariate — would produce more appropriate within-role comparisons.

6. Additional Figures

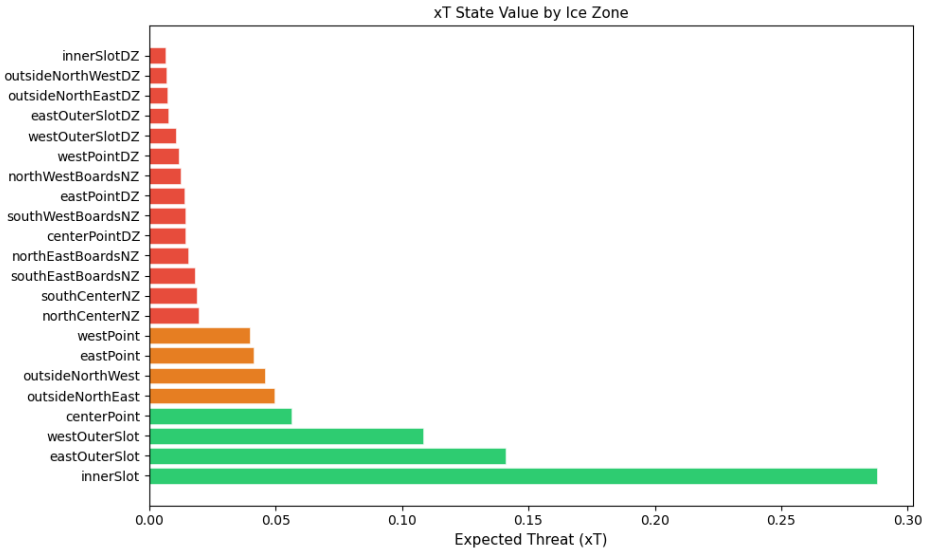


Figure 3: Expected Threat (xT) by ice zone, estimated from the full dataset. Green zones carry substantial scoring probability; orange zones are intermediate; red zones are low. The ordering follows hockey intuition: the inner slot is the highest-value zone, followed by outer slot areas, points, and neutral/defensive zones

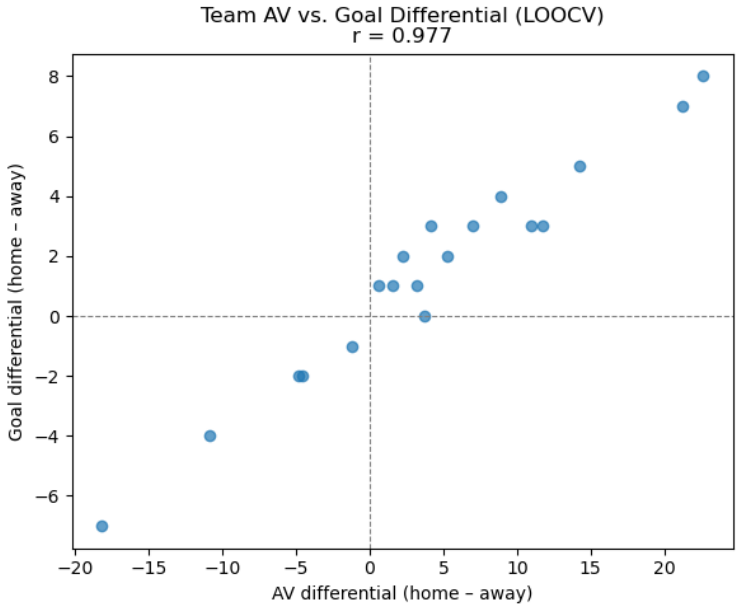


Figure 4: Team AV differential vs. goal differential per game (LOOCV). The strong positive correlation ($r = 0.977$) confirms internal consistency. Note that this correlation is structurally elevated because goals are a direct model input; it should be interpreted as a coherence check rather than a test of predictive validity.